SMART RECIPE

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

in

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

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Declaration

The Project Report entitled **"Smart Recipe"** is a record of Bonafide work of **Sai Lasya Grandhi – 2520030538, Samshritha Ramisetty – 2520030045, Laxmi sri Gundla – 2520030264** submitted in partial fulfillment for the award of B. Tech in Computer Science and Engineering. The results embodied in this report have not been copied from any other departments/University/Institute.

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CERTIFICATE

This is certify that the mini project based report entitled **"Smart Recipe"** is a bonafide work done and submitted by **Sai Lasya Grandhi – 2520030538, Samshritha Ramisetty – 2520030045, Laxmi sri Gundla – 2520030264** in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** in Department of Computer Science Engineering, during the academic year **2026-27.**

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Signature of the HOD

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ACKNOWLEDGEMENT

Description in Times New Roman font 12 size with two to three paragraphs thanking guide Dr. S. Vinay Kumar, Associate Professor, Dept. of CSE and Dr. P. Venkateshwara Rao, Associate Professor, HOD CSE

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Abstract

The Smart Recipe – DSA Based Recipe Recommendation System is a web-based application designed to help users discover suitable recipes based on the ingredients available with them. In many situations, users have several ingredients at home but are unsure about what dishes can be prepared using them. Traditional recipe-search methods generally require users to search manually using exact recipe names or individual ingredients. This can be inconvenient when the user does not know the recipe name or enters an ingredient with a spelling mistake. The proposed system addresses this problem by providing an ingredient-based recipe recommendation platform.

The main focus of the project is the application of Data Structures and Algorithms to improve the recipe-search and recommendation process. The system uses pattern matching to identify exact ingredient matches between user input and recipe data. It also incorporates fuzzy search using the Levenshtein Distance algorithm so that the system can identify approximately matching words when users make spelling mistakes. These algorithms allow the system to provide useful results even when the input is not perfectly formatted or spelled.

The project also uses a similarity algorithm to compare the ingredients entered by the user with the ingredients required for available recipes. Jaccard Similarity can be used to determine the degree of overlap between two ingredient sets. Based on the calculated matching and similarity scores, recipes can be ranked and presented to the user. This demonstrates how DSA concepts can be applied to a practical real-world problem rather than being implemented only as theoretical programs.

The system is designed as a simple, responsive and user-friendly web application. The frontend provides an ingredient search interface, recipe cards, recipe details and recommendation results. The project is being developed incrementally, beginning with repository creation and frontend development, followed by implementation of pattern matching, fuzzy search and similarity algorithms. The final objective is to provide a simple recipe-discovery system while demonstrating the practical use of DSA concepts in web application development.

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1. INTRODUCTION

1.1 Overview

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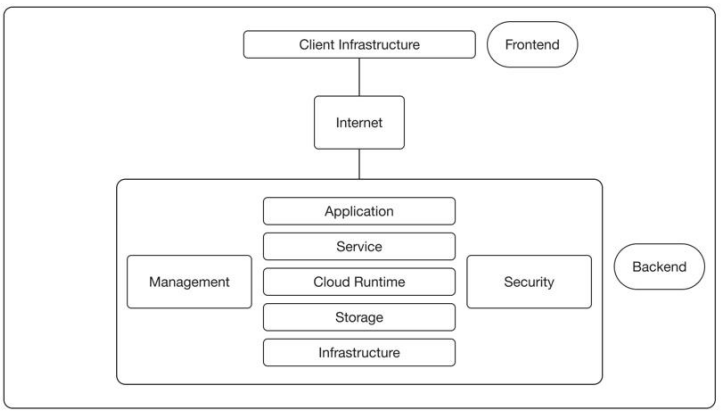


Fig. 1.1 Architecture of Cloud

1.2 Problem Statemen

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1. INTRODUCTION

1.1 Overview

Smart Recipe is a web-based recipe recommendation system developed to help users find recipes according to the ingredients they already have. Instead of searching for a recipe by its exact name, the user can enter available ingredients such as tomato, onion, rice, potato or garlic. The system processes these ingredients and searches the recipe dataset for suitable matches. This makes recipe discovery easier and more convenient for users who want to prepare food using ingredients that are already available.

The project mainly focuses on applying Data Structures and Algorithms to a practical recipe-search problem. Pattern matching is used to identify whether a particular ingredient exists within a recipe. Fuzzy search is used when the user enters an ingredient incorrectly or with a spelling variation. Similarity calculation is then used to determine how closely a recipe matches the ingredients provided by the user. These operations form the core computational logic of the application.

The system consists of a recipe dataset containing recipe names, ingredients, descriptions, cooking time, difficulty level and preparation instructions. When the user enters ingredients, the system processes the input and compares it with the available recipe information. Exact matches are identified first, while unmatched inputs can be processed using fuzzy search. The resulting recipes can then be ranked according to the number of matching ingredients and their similarity scores.

The project is being developed as a DSA-oriented academic project, where the primary goal is not only to provide a functional recipe website but also to demonstrate algorithmic thinking. The project provides an opportunity to understand how pattern matching, string comparison, set operations and similarity calculations can be combined to solve a real-world recommendation problem. The system therefore connects theoretical DSA concepts with practical web development.

1.2 Problem Statement

Finding recipes based on ingredients available at home can be difficult for users. Most existing recipe platforms are designed around recipe names, food categories or manually selected ingredients. A user may have ingredients such as tomato, onion and rice but may not know which recipes can be prepared using them. Searching each ingredient individually can also produce many irrelevant results and requires additional effort from the user.

Another problem occurs when users enter ingredient names incorrectly. For example, a user may enter “tomtao” instead of “tomato” or “onoin” instead of “onion”. A simple exact-search system would fail to identify these inputs even though the intended ingredients are obvious. Therefore, the system needs a mechanism that can identify approximate matches between the user's input and the ingredient names stored in the recipe dataset.

The system also needs to determine which recipe is the most suitable when several recipes contain some of the same ingredients. Simply checking whether an ingredient exists is not sufficient for effective recommendation. A recipe containing three of the user's four ingredients should generally receive a higher recommendation than a recipe containing only one matching ingredient. Therefore, a similarity calculation is required to compare ingredient sets and rank recipes.

The problem addressed by this project is therefore the development of a recipe recommendation system that can accept user ingredients, perform exact pattern matching, handle spelling variations using fuzzy search, calculate recipe similarity and provide relevant recommendations. The system aims to demonstrate how these DSA techniques can work together in a practical application.

1.3 Scope of the Project

The scope of the Smart Recipe project includes the development of a web-based interface through which users can enter ingredients available to them. The system will process the entered ingredients and search a predefined recipe dataset. The initial implementation focuses on a manageable collection of recipes so that the DSA algorithms can be clearly demonstrated and tested. Each recipe can contain information such as name, ingredients, description, cooking time, difficulty and preparation instructions.

The project covers the implementation of exact pattern matching for identifying ingredients within recipes. It also includes fuzzy searching so that small spelling mistakes can be detected and corrected or matched with the closest available ingredient. The Levenshtein Distance algorithm can be used to measure the number of character-level changes required to transform one word into another. This functionality improves the flexibility of the search system.

The project also covers similarity-based recipe recommendation. The ingredients supplied by the user can be represented as a set and compared with the ingredient set of each recipe. Jaccard Similarity can be used to calculate the overlap between the two sets. Recipes can then be sorted according to their calculated scores, allowing the user to see the recipes that most closely match the available ingredients.

The present scope is focused on the partial implementation required for the academic project review. It includes repository creation, frontend development, recipe dataset creation and implementation of pattern matching, fuzzy search and similarity algorithms. Future versions can extend the scope by adding a larger recipe database, user accounts, personalized recommendations, nutritional information and advanced search features.

1.4 Objectives

The primary objective of Smart Recipe is to develop an easy-to-use recipe recommendation system that suggests recipes based on ingredients available to the user. The system should allow users to enter multiple ingredients and receive a list of recipes that match those ingredients. The interface should be simple enough for users to understand without requiring technical knowledge.

A major objective of the project is to implement pattern matching as a practical DSA technique. The system should be capable of checking whether an ingredient entered by the user occurs in the ingredient information of a recipe. This provides the foundation for identifying recipes that contain the required ingredients.

Another objective is to implement fuzzy search using an algorithm such as Levenshtein Distance. This allows the system to handle common spelling mistakes and variations in user input. The project also aims to implement a similarity algorithm such as Jaccard Similarity to calculate how closely the user's ingredients correspond to the ingredients of each recipe.

The final objective is to combine these algorithms into a working recommendation workflow. The system should process the user's input, identify exact and approximate matches, calculate similarity scores and rank suitable recipes. Through this implementation, the project demonstrates the practical application of DSA concepts in a web-based system.

2. SYSTEM REQUIREMENTS

2.1 Functional Requirements

The Smart Recipe system should provide an interface for entering ingredients available to the user. The user should be able to enter one or multiple ingredients separated by commas or through individual ingredient fields. The system should validate the input and provide an appropriate message if no ingredients are entered. The search operation should begin when the user selects the recipe-search option.

The system should perform exact pattern matching between the entered ingredients and recipe ingredients. When an exact match is found, the corresponding recipe should receive a matching result. The system should also identify ingredients that do not match exactly and pass them through the fuzzy-search process. This allows the application to handle both correctly and incorrectly entered ingredient names.

The fuzzy-search functionality should calculate the similarity between an unmatched input and known ingredient names. When the distance is within an acceptable threshold, the system should consider the words as a possible match. After ingredient matching, the system should calculate the similarity between the user's ingredient set and the ingredients of each recipe.

The application should display the resulting recipes in an understandable format. Each recipe can display its name, description, matching ingredients, similarity score, cooking time and difficulty level. The user should also be able to open a recipe to view its complete ingredients and preparation instructions.

2.2 Non-Functional Requirements

The system should provide a simple and user-friendly interface. Users should be able to understand how to enter ingredients and start the search without requiring technical knowledge. Navigation elements, buttons, input fields and recipe cards should be clearly visible and consistently designed.

The system should also provide reasonable performance for the recipe dataset used during the project. The algorithms should process the available recipes without unnecessary delays. Since the project is focused on DSA implementation, the performance of pattern matching, fuzzy search and similarity calculations should be considered when designing the implementation.

The application should be responsive so that the interface can adapt to different screen sizes. The website should be usable on desktop and mobile screens. The design should maintain readable text, properly sized buttons and organized recipe cards across different screen dimensions.

The system should also be maintainable and understandable. The DSA algorithms should be implemented in separate functions or modules with meaningful names and comments. The project repository should have an organized folder structure so that team members and evaluators can easily understand the different parts of the application.

3. SYSTEM DESIGN

3.1 High-Level Architecture

The Smart Recipe system can be designed using a layered web application structure. The frontend layer provides the user interface through which ingredients are entered and recipe results are displayed. The application logic layer handles input processing, pattern matching, fuzzy searching and similarity calculations. The data layer contains the recipe information used by the algorithms.

The overall process begins when the user enters available ingredients through the frontend. The input is sent to the application logic, where it is cleaned and divided into individual ingredients. The system first attempts to identify exact matches using the pattern-matching algorithm. Ingredients that cannot be matched directly are passed to the fuzzy-search component.

After the ingredients are processed, the system compares the resulting ingredient set with the ingredients of available recipes. Similarity values are calculated using the selected similarity algorithm. The recipes can then be sorted according to their scores and returned to the frontend for display.

The architecture therefore follows a simple flow from user input to processing and finally to recommendation. This design separates the user interface from the algorithmic processing and recipe data. Such separation makes the project easier to understand, test and modify as additional features are introduced.

3.2 Description of Each Layer

The presentation layer is responsible for displaying the Smart Recipe interface. It contains the navigation bar, ingredient input section, search button, recipe cards, recommendation results and recipe-detail interface. The frontend is designed to make the interaction simple and visually clear for the user.

The application logic layer contains the main processing operations. It receives the ingredients entered by the user and performs preprocessing before applying the DSA algorithms. Pattern matching is responsible for exact matches, fuzzy search handles approximate matches and similarity calculation determines the degree of correspondence between ingredient sets.

The data layer contains the recipe information required by the system. Each recipe can be represented as a structured object containing fields such as recipe name, ingredient list, description, cooking time, difficulty and instructions. For the partial implementation, the recipe data can be maintained locally rather than relying on an external database.

The Node.js environment can be used to run the project and provide the development/server environment for the application. The frontend and application logic can be organized into separate files or modules. This structure provides a clear separation of responsibilities and allows the DSA implementation to be developed independently from the visual interface.

4. IMPLEMENTATION

4.1 Module-Wise Implementation

The Smart Recipe application is divided into several functional modules. The user interface module handles ingredient input and displays recipe recommendations. The recipe-data module maintains the available recipes and their associated information. The search module processes user input and connects it with the implemented DSA algorithms.

The pattern-matching module is responsible for identifying exact ingredient matches. A Naive Pattern Matching approach can compare the input pattern with the recipe ingredient text or individual ingredient values. The algorithm checks possible positions or values until a matching pattern is found. This demonstrates the use of string-search techniques in a practical application.

The fuzzy-search module handles ingredients that do not produce an exact match. Levenshtein Distance can be implemented using dynamic programming to calculate the minimum number of insertions, deletions or substitutions required to transform one string into another. A threshold can then be used to determine whether an input is sufficiently close to a known ingredient.

The similarity module compares the user's ingredients with the ingredients associated with each recipe. Jaccard Similarity can be used to determine the ratio between common ingredients and the total unique ingredients. The resulting score can be used to rank recipes and display the most relevant recommendations first.

4.2 Front-End Logic and UI Rendering

The frontend provides the primary interaction point for the Smart Recipe system. It contains a search area where users can enter ingredients and a results section where recommended recipes are displayed. Recipe cards can present important information such as recipe name, matching ingredients, match percentage, cooking time and difficulty.

When the user submits the search, the frontend collects the entered values and performs basic preprocessing. This can include converting the input to lowercase, removing unnecessary spaces and separating multiple ingredients. The processed values are then passed to the appropriate search and recommendation functions.

The frontend also handles the rendering of recommendation results. After the algorithms produce the results, JavaScript can dynamically generate recipe cards and insert them into the result section. If no suitable recipe is found, the interface should display a clear message instead of leaving the results area empty.

The user interface can also provide additional interactions such as viewing recipe details, clearing the search and displaying fuzzy-match suggestions. The design should remain consistent with the Smart Recipe concept and should make the DSA-driven recommendation process easy to demonstrate during the project review.

4.3 Pattern Matching Implementation

Pattern matching is implemented to identify exact ingredient matches within recipe information. The project can use the Naive Pattern Matching algorithm because it is simple to understand and provides a clear demonstration of a fundamental string-search technique. The algorithm compares the search pattern with possible positions in the text until a match is found or all possible positions have been checked.

For example, if the user enters “tomato”, the system can search recipe ingredients for the same pattern. If a recipe contains tomato, the algorithm identifies it as an exact match. The matching recipe can then be included in the recommendation process.

The algorithm can be implemented manually rather than using a built-in search function as the primary DSA implementation. This is important because the purpose of the project review is to demonstrate the algorithm itself. The implementation should therefore contain a dedicated function with clear comments describing how each comparison is performed.

The pattern-matching result can be combined with the other algorithms. Exact matches can receive a higher priority during recommendation, while unmatched ingredients can be passed to the fuzzy-search stage. In this way, pattern matching becomes the first stage of the Smart Recipe recommendation pipeline.

4.4 Fuzzy Search Implementation

Fuzzy search is implemented to improve the system's ability to understand user input containing spelling mistakes. The project can use Levenshtein Distance to compare the entered ingredient with known ingredient names. The algorithm calculates the minimum number of character-level operations needed to transform one string into another.

For example, if the user enters “tomtao” instead of “tomato”, the system can compare the two words and calculate their edit distance. If the distance is within the selected threshold, the system can identify “tomato” as the likely intended ingredient. This allows recipes containing tomato to be considered during recommendation.

The implementation can use a dynamic-programming table to store intermediate results. Each cell represents the minimum edit distance between portions of the two strings. This prevents repeated calculations and provides an efficient method for calculating the final distance.

The fuzzy-search result can be displayed to the user as a suggestion or correction. For example, the interface can show “tomtao → tomato”. This makes the system more understandable and also provides visible evidence during the project review that the fuzzy-search algorithm is actually being used.

4.5 Similarity Algorithm Implementation

The similarity module is responsible for determining how closely a recipe matches the ingredients entered by the user. For the initial implementation, Jaccard Similarity is suitable because both the user's ingredients and recipe ingredients can be represented as sets. The algorithm compares their intersection with their union.

For example, if the user's ingredients are tomato, onion and rice, while a recipe contains tomato, onion, rice and garlic, the intersection contains three common ingredients. The union contains four unique ingredients. Therefore, the Jaccard Similarity is calculated as 3 divided by 4, producing a similarity score of 75 percent.

The calculated score can be used to rank the available recipes. A recipe with a higher similarity value is considered a better match for the user's available ingredients. The system can sort the results from the highest score to the lowest score before displaying them on the website.

The similarity algorithm therefore converts ingredient matching into a measurable value. This makes the recommendation process more systematic rather than simply displaying every recipe that contains one matching ingredient. It also provides a clear DSA component that can be demonstrated during the review.

5. RESULTS

5.1 System Output

The partial implementation of the Smart Recipe system is expected to accept ingredient input and produce relevant recipe recommendations. The user can enter multiple ingredients and initiate the search. The system processes the input and identifies recipes containing matching ingredients.

Exact ingredient matches are identified through pattern matching. When a user enters a correctly spelled ingredient, the corresponding recipes can be identified directly. This demonstrates that the pattern-matching module is connected to the recipe-search functionality.

For incorrectly spelled ingredients, the fuzzy-search module can identify possible intended ingredients. For example, an input such as “tomtao” can be compared with “tomato” using Levenshtein Distance. If the calculated distance satisfies the selected threshold, the system can use tomato for further recipe matching.

The final output consists of ranked recipe recommendations. Each recommendation can display the recipe name, matching ingredients and similarity score. This provides a visible demonstration of the complete workflow from user input through DSA processing to recipe recommendation.

6. CHALLENGES AND LIMITATIONS

6.1 Issues Faced During Development

One of the main challenges in the project is connecting theoretical DSA algorithms with a practical web application. Implementing an algorithm separately is relatively straightforward, but integrating its output into a frontend recommendation workflow requires additional processing. The input format, recipe structure and algorithm outputs must be designed to work together.

Another challenge is handling user input correctly. Users may enter ingredients using different capitalization, additional spaces or spelling mistakes. The system therefore needs preprocessing before applying the algorithms. Without proper preprocessing, the algorithms may produce incorrect or incomplete results.

Managing recipe data is another challenge. The recommendation quality depends on the ingredients stored for each recipe. A small or incomplete dataset can limit the number of useful recommendations. Therefore, the recipe data needs to be structured consistently so that the algorithms can compare ingredients correctly.

Another challenge is making the project understandable during demonstration. Since the project is focused on DSA, the implementation should clearly show how each algorithm contributes to the final result. The team needs to explain the input, processing steps, algorithm output and final recommendation in a simple manner.

6.2 Solutions Applied

To handle input inconsistencies, the system can preprocess all ingredient names by converting them to lowercase and removing unnecessary spaces. Multiple ingredients can also be separated into individual values before processing. This provides a consistent input format for the algorithms.

For spelling mistakes, the fuzzy-search module provides a solution by comparing unmatched inputs with known ingredient names. Levenshtein Distance can identify close matches based on character-level differences. A threshold can prevent unrelated words from being treated as valid matches.

To improve recommendation quality, the similarity module calculates the overlap between user ingredients and recipe ingredients. Instead of returning recipes randomly, the system can sort them according to their similarity scores. This allows recipes with more matching ingredients to appear earlier in the results.

The project can also be organized into separate modules for each algorithm. Keeping pattern matching, fuzzy search, similarity calculation, recipe data and frontend logic separated makes debugging easier. It also allows each algorithm to be tested independently before being integrated into the complete system.

6.3 Limitations of the System

The initial Smart Recipe implementation depends on a predefined recipe dataset. If a particular recipe or ingredient is not present in the dataset, the system may not be able to recommend it. Therefore, the quality and coverage of the recommendations are limited by the available recipe information.

The fuzzy-search algorithm can also produce incorrect suggestions when two ingredient names are sufficiently similar in terms of characters. A spelling correction based only on edit distance does not understand the actual meaning of the word. Therefore, the threshold must be selected carefully.

The current similarity calculation mainly considers ingredient overlap. It does not necessarily consider quantities, cooking preferences, nutritional requirements or ingredient importance. Two recipes can receive similar scores even though one may require a significantly larger quantity of an unavailable ingredient.

The system also does not currently provide personalized recommendations based on user history, dietary preferences or previous ratings. These features can be added in future versions. The present implementation is primarily intended to demonstrate the DSA algorithms and their integration into a basic recipe recommendation system.

7. CONCLUSION AND FUTURE WORK

7.1 Conclusion

The Smart Recipe – DSA Based Recipe Recommendation System demonstrates how Data Structures and Algorithms can be applied to a practical web-based problem. The system allows users to search for recipes using ingredients that are already available to them. This provides a more direct approach to recipe discovery than requiring users to know the exact name of the dish they want to prepare.

The project implements pattern matching to identify exact ingredient matches. It also uses fuzzy search through Levenshtein Distance to handle spelling variations and approximate input. These algorithms improve the flexibility of the recipe-search process and demonstrate the practical application of string-processing techniques.

Jaccard Similarity provides a method for comparing the user's ingredient set with recipe ingredient sets. The similarity score allows the system to rank recipes based on how closely they match the user's available ingredients. This transforms the basic search functionality into a recommendation-oriented system.

The project also provides valuable experience in combining DSA concepts with web development. The partial implementation covers repository creation, frontend development, recipe data handling and the core algorithms required for the project review. The system provides a foundation that can be expanded with additional features in future development stages.

7.2 Future Work

One possible future enhancement is to expand the recipe dataset significantly. A larger dataset would provide users with more diverse recommendations and improve the usefulness of the system. Recipe information could also include nutritional values, preparation time, serving size and detailed ingredient quantities.

The recommendation algorithm can also be improved in future versions. Instead of relying only on ingredient overlap, the system could consider ingredient quantities, recipe categories, dietary preferences and user ratings. More advanced similarity and recommendation techniques could be incorporated to improve ranking accuracy.

The system could also include user accounts and personalized recipe history. Users could save their favorite recipes, view previously searched ingredients and receive recommendations based on their previous activity. Dietary filters such as vegetarian, vegan or high-protein options could also be added.

Another future enhancement is the integration of a larger backend database and external recipe services. This would allow recipes to be updated dynamically instead of maintaining only a fixed local dataset. The system could eventually become a complete personalized recipe recommendation platform while retaining the DSA-based search and similarity components.

7.3 Skills Learned During Development

The project provides practical experience in Data Structures and Algorithms, particularly pattern matching, string comparison and similarity calculations. Implementing these algorithms manually helps the team understand how theoretical concepts can be translated into executable program logic. The project also provides an opportunity to analyze algorithm behavior and understand the importance of efficient processing.

The team also gains experience in web application development. Creating the Smart Recipe interface involves working with frontend structure, styling, user interaction and dynamic rendering of recipe information. Integrating the frontend with the algorithmic logic helps demonstrate how different areas of software development work together.

Another important skill developed through the project is problem-solving. The team must consider how to process inconsistent user input, organize recipe data, handle spelling mistakes and rank multiple possible results. These challenges require logical thinking and systematic testing.

Finally, the project provides experience in collaborative software development and repository management. Maintaining the project in a GitHub repository allows team members to organize code and track development. The project also improves presentation and technical communication skills because the team must explain the architecture, algorithms, implementation and results during project reviews.

REFERENCES

The references section should contain the **actual books, tutorials, documentation sites, libraries and other resources that your team genuinely used** during development. The sample documentation also places references at the end of the report and includes books, tutorials, APIs and documentation sites. Project Documentation format 20…

You can structure them like:

1. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest and Clifford Stein, *Introduction to Algorithms*, MIT Press.
2. Robert Lafore, *Data Structures and Algorithms in Java*, Sams Publishing.
3. Node.js Documentation — official documentation used for Node.js development.
4. MDN Web Docs — HTML, CSS and JavaScript documentation.
5. GitHub Documentation — repository creation and version-control documentation.
6. Documentation and learning resources consulted for string matching, Levenshtein Distance and Jaccard Similarity during development.

Important: Replace generic references with the **exact resources your team actually used** before submitting the report.